Demographic and Fiscal Modeling for Environmental Planning

New tools from NSF-funded projects

Tuesday, April 22, 2025

Gov. James J Florio Special Events Forum

Edward J Bloustein School of Planning and Public Policy

Rutgers University New Brunswick

33 Livingston Ave, New Brunswick, NJ

Sponsored by the New Jersey Chapter of the American Planning Association and the Bloustein School

This work is supported by National Science Foundation Grants CMMI-2053637 and ICER-2103754

Program

- **3:30 pm INTRODUCTION AND WELCOME** Prof. David Listokin, Blaustein School, Rutgers University, and Faculty Liasion on APA New Jersey Executive Committee
- **3:35 pm DEMOGRAPHIC, FISCAL, and ENVIRONMENTAL MODELING RESEARCH**
  - Overview of NSF Project Prof, Clinton Andrews, AICP, Blaustein School, Rutgers University (moderator)
  - New Jersey's Floodi,ig Exposure and Implications (Online NJADAPT Portal) Pritpal Bamhra} Center for Urban Policy Research, Rutgers University
  - **New** Jersey Resident Perspectives on Flood **Risk** and Response (Eagleton Survey) Garin Bulger. Center for Urban Policy Research, Rutgers University
  - Residential Demographic Multipliers for lrnpact Modeling (Online Update for 50 U.S. States David Listokin and **Umer Farooq**, Center for Urban Policy Research, Rutgers University
  - Local Fiscal Impacts of Flood Events **(New** Online Fiscal Impact Analys!,s Model) Jungh'oon Lee, Center for Urban Policy Research, Rutgers University
- **PERSPECTIVES ON THE RESEARCH**
  - J. Creigh Rahenkamp, NJPP, Creigh Rahenkamp & Associates, LLC, APA New Jersey Housing Chair (moderator)
  - Stan Slachetka, FAICP, PP, Special Projects Planning Leader, Pennoni
  - Donna Rendeiro, Former Executive Director. Office of Planning Advocacy and the State Planning Commission
  - Robert Goldsmith, Partner/Co-Chair Redevelopment and Land Use Department, Greenbaum, Rowe, Smith & Davis LLP
- **5:10 pm DISCUSSION**
- **5:30 pm LIGHT REFRESHMENTS (Forum Lobby)**

National Science Foundation grants

Modeling Intergovernmental Fiscal Impacts of Coastal Hazards (2053637)

- Adapt fiscal impact analysis to study coastal hazards
- Update demographic multipliers used in impact studies
- Understand public perceptions of coastal hazard management alternatives
- Clarify intergovernmental fiscal flows and their drivers

Megalopolitan Coastal Transformation Hub (2103754)

- Strengthen scientific basis to develop effective, evidence-based responses to coastal climate change risks, with a primary focus in the Philadelphia-New Jersey-New York City region
- Produce great science, train future workforce, learn from partners

https://cupr.rutgers.edu/products/calculators-and-dashboards/

Flooding Exposure Impacts on Municipal Public Finance in New Jersey

Pritpal Bamhrah, **Umer Farooq**, and David Listokin

04/22/2025

Contents

Overview of NJ Adapt, MOD IV data, and environmental challenges

Analytic strategy and Hoboken, Hudson County Illustration

New Jersey county and statewide summary impacts by flooding scenarios

Intersection of NJ flood exposure to Statewide Property Buyouts, Construction Activity and municipal bond ratings

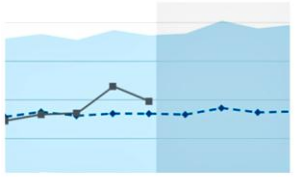
Accomplishments, limitations and future analytic enhancements

Overview of NJ Adapt

Link - https://njclimateresourcecenter.rutgers.edu/nj-adapt/

NJ Adapt is a suite of tools developed by Rutgers University for climate adaptation planning in New Jersey to assist planners, community leaders, businesses, and residents to understand and adapt to the impacts of climate change on people, assets, and communities in New Jersey.

Tools within NJ Adapt



Climate Dashboard

New Jersey climate trends in moderate and high emissions scenarios



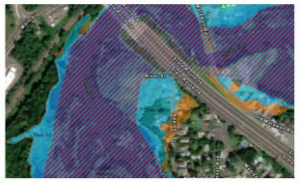
Climate Planning Tool

A guide to using coastal flooding data in climate change planning



Climate Snapshots

Climate risks summarized by municipality, county and statewide



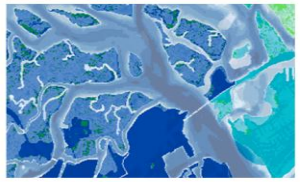
Local Planning Navigator

A decision-support tool for building community resilience



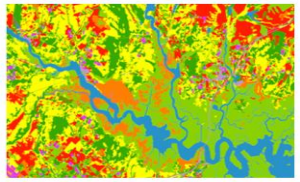
NJ ResTOrS

New Jersey Restoration Tool Organization Suite (NJResTOrS)



NJ FloodMapper

An interactive flood exposure data mapping tool



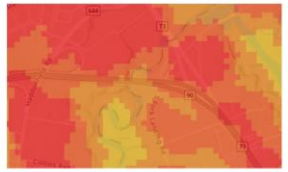
NJ Forest Adapt

A data mapping tool for forest management



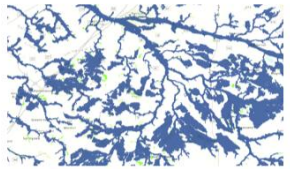
NJ HazAdapt

Data and resources for hazard mitigation planning



NJ Public Health Adapt

Climate planning for improved health outcomes



Rutgers Inland Design Flood Elevation

FEMA 1% Chance Annual Flood +3 feet

Disclaimer

- The data, maps, and information provided should be used *only* as a screening-level tool. As with all remotely sensed data, all features should be verified with a site visit.
- These tools should be used *strictly* for planning reference - and not for navigation, permitting, or other legal purposes.

NJ Parcel Level Tax Data - MOD IV

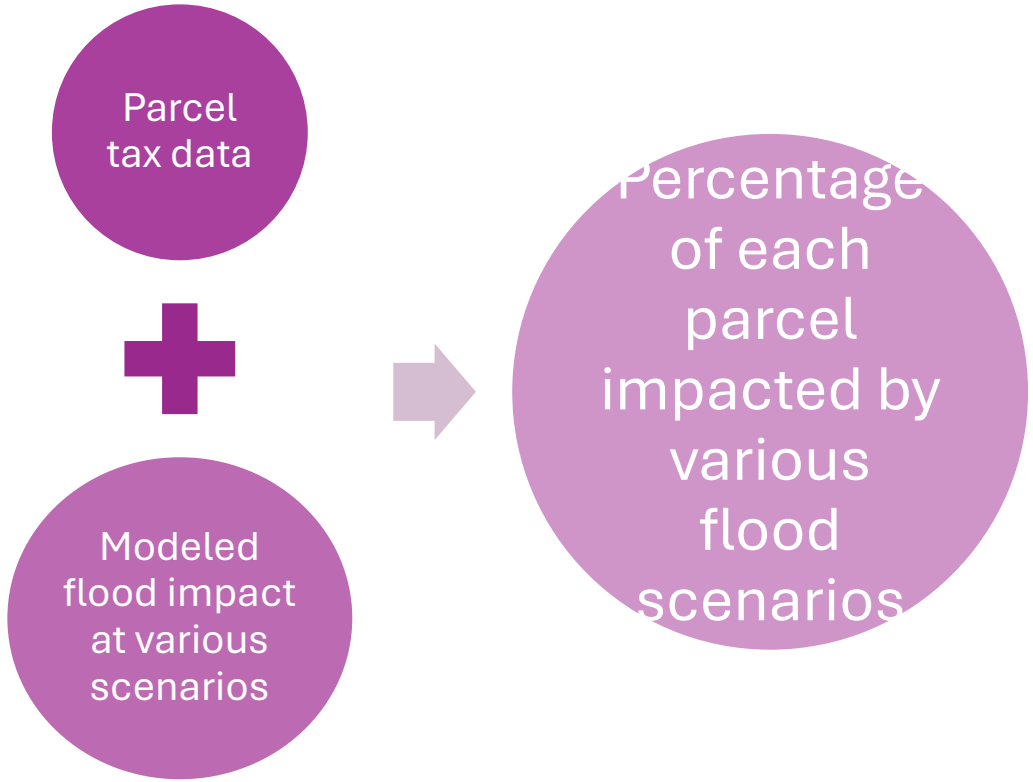
MOD IV is a NJ government data system designed for the uniform preparation, maintenance, presentation, and storage of property tax information in New Jersey. It contains property assessment (valuation) records and generates tax lists, accounting for all real property parcels and their taxable values.

MOD IV Flood Analysis: Located within NJ HazAdapt, this dataset provides potential flood analysis for *each* tax parcel in New Jersey. It provides the percentage of each parcel flooded by:

- Inundation above Mean Higher High Water (MHHW) for Coastal Flooding - in 1ft increments
- Federal Emergency Management Agency (FEMA) Flood Zones

The current analysis uses **2022 MOD IV data**.

NJ Adapt MOD-IV Parcel Flood Analysis

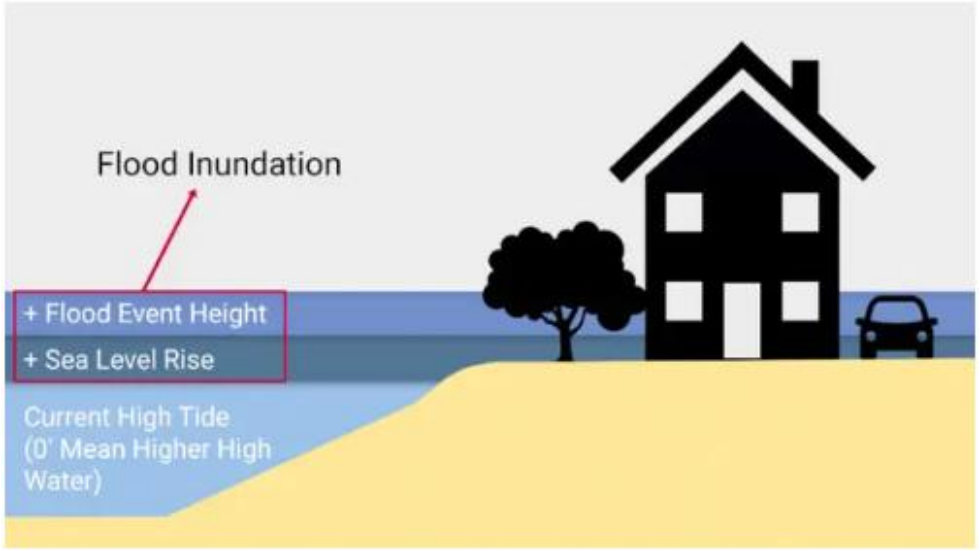


Analysis considers both property assessed value - $ amount of the property value (both land and improvements) recorded on the official tax record, which may differ from the property market value or “equalized” property value.

Environmental Challenges: Flooding Scenarios

- **Coastal Flooding**: Tidal flooding, rising sea levels and storm events cause coastal flooding. NJ Adapt provides the inundation values above MHHW levels.
  - The amount of nuisance flooding some parts of New Jersey are currently experiencing at highest high-tide is 2ft
  - NJ DEP recommends planning for year 2100 when the corresponding sea-level rise will be 5ft

- **Inland Flooding**: Inland flooding occurs along rivers and streams when intense rain events cause them to overtop their banks. The Federal Emergency Management Agency (FEMA) models these flood hazards for much of the state.



Environmental Challenges: Flooding Scenarios

- **Environmental Challenge Scenarios**: Analyzing flooding impacts under various sea-level rise and inland flood scenarios.
- **Scenarios Included**:
  - **Coastal Flooding**: Assessments at 2 ft and 5 ft flooding (inundation above MHHW)
  - **Inland Flooding**: Includes FEMA 100-year and 500-year flooding estimates.
  - **Combined Scenario**: 5ft flooding above MHHW + FEMA 500 Year flooding condition estimate.
  - The flooded parcels included in each scenario have more than 50% of the area flooded.

Five Flooding Scenarios – more than 50% flooded
1. Coastal Scenario 2ft
2. Coastal Scenario 5ft
3. Combined Scenario 5ft+ FEMA 500
4. Inland Scenario FEMA 100
5. Inland Scenario FEMA 500

- **Rationale**:
  - FEMA data was chosen in scenario 3 (combined scenario) to represent both inland and coastal flooding impacts accurately.
  - A range of flooding scenarios provide a comprehensive view of potential flooding risks under different conditions.

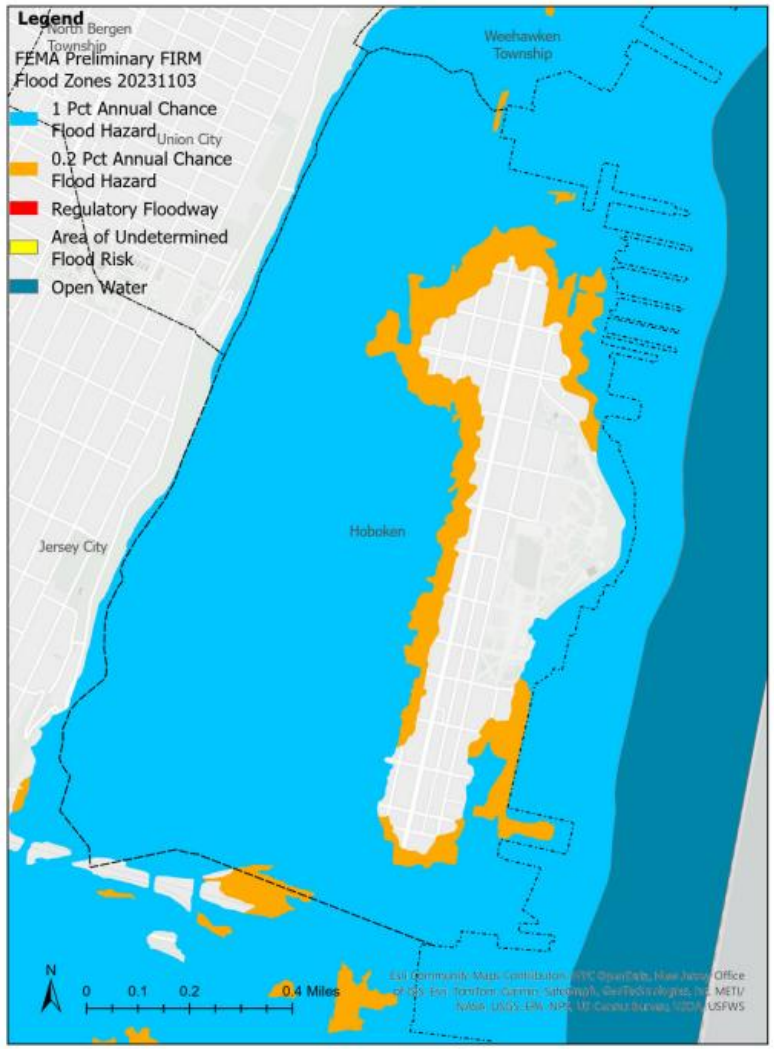
Analytic Strategy

- The analysis uses data from MOD-IV intersected for different flood inundation scenarios and FEMA Flood zones. Analyzes impacts on:
    - number of parcels,
    - acreage,
    - property values, and
    - annual property taxes
- This study provides critical information for both municipal finance and climate adaptation

Data Aggregation Approach:

- Starting from an individual parcel level, data are aggregated to the *Community-scale* (Sub-County/ Municipalities), ensuring data accuracy and integrity. (New Jersey has 564 Municipalities.)
- Extended aggregation to the *County scale* for broader analysis (New Jersey has 21 counties) as well *Statewide*.

Parcel Analysis at Community Level – Hoboken Example



Example – Hoboken at Hudson County, NJ

Land Area impacted under FEMA Flood Zones

Parcels exposed to flooding beyond 50% (FEMA 500year+5ft above MHHW, combined scenario)

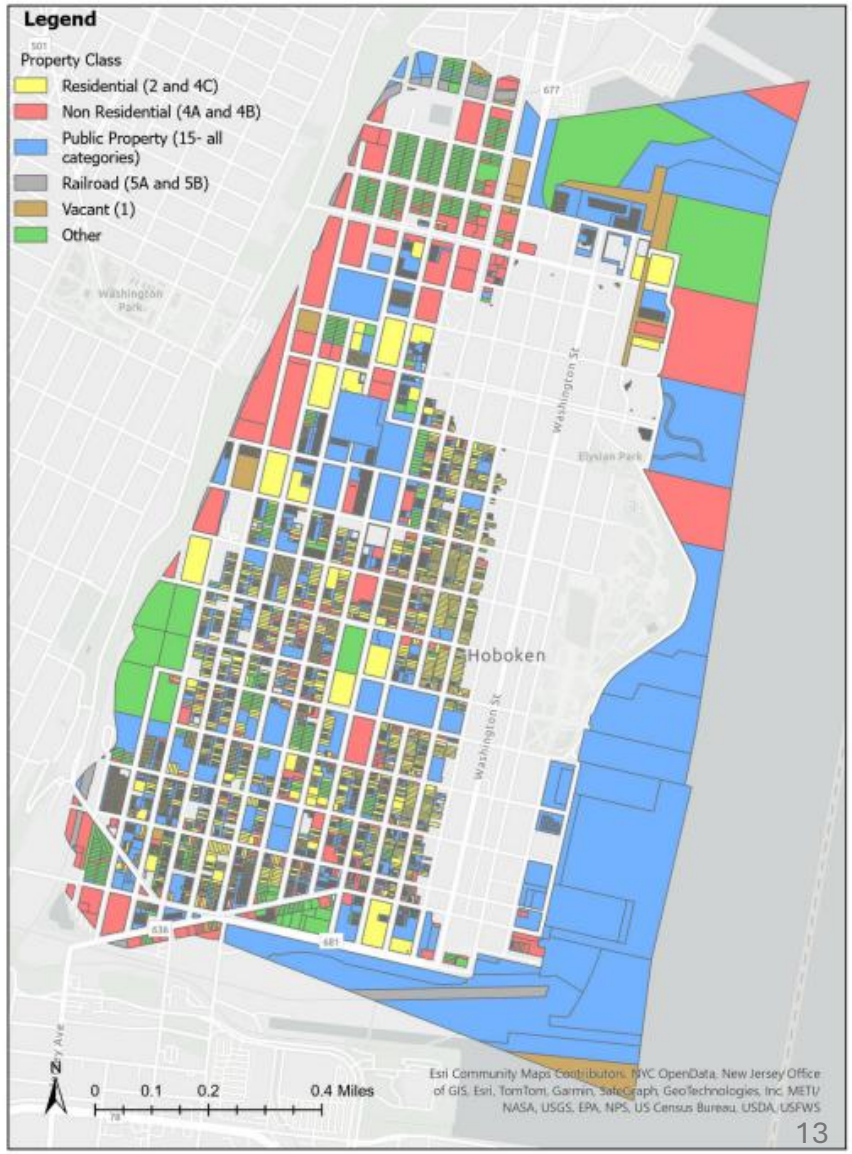
Hoboken confronts extensive flooding challenges.



Parcel Analysis at Community Level – Hoboken Example

Once parcels are identified, MOD IV data helps analyze flooding exposure by property class and total assessed (land and improvement) values.

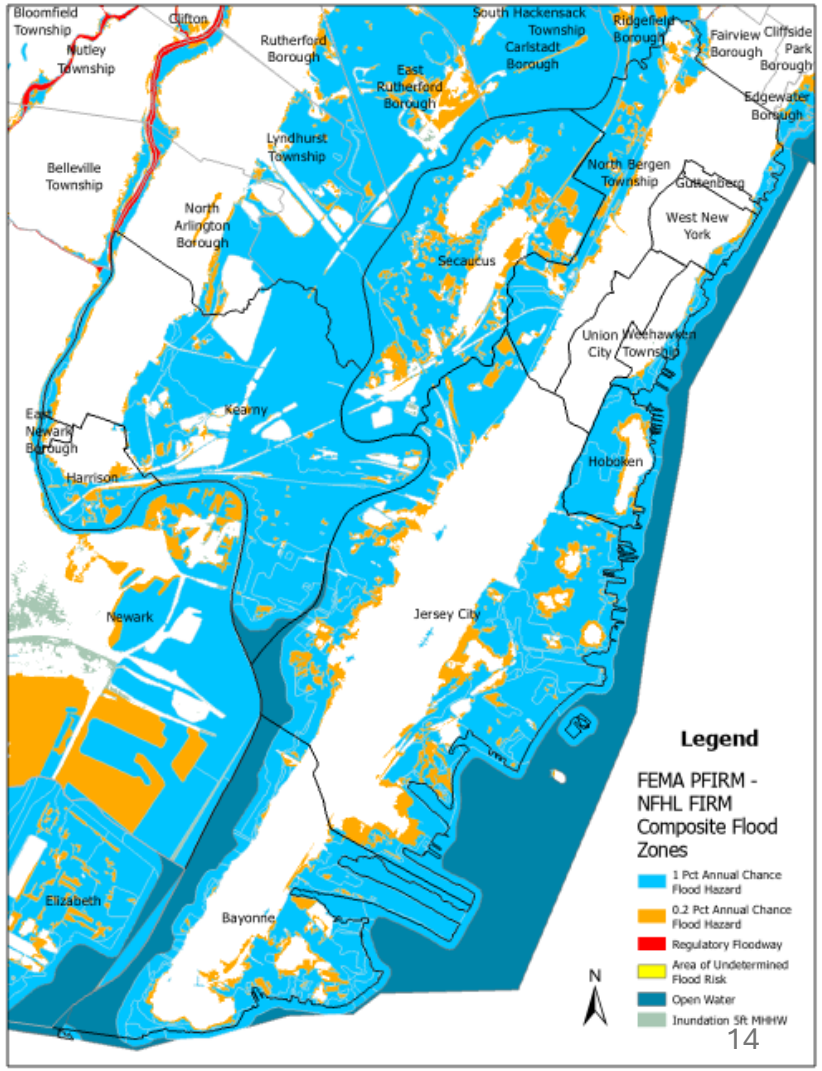
Property Class	No. of Parcels (%)	Assessed Value (Land Value + Imp Value)
Residential (2 and 4C)	10823 (84%)	$ 5,941,211,100
Non-Residential (4A and 4B)	495 (4%)	$ 989,459,400
Public Property (15 - all categories)	937 (7%)	$ 1,636,622,800
Railroad (5A and 5B)	21 (0%)	$ 6,232,900
Vacant (1)	68 (1%)	$ 21,806,200
Other	531 (4%)	$ -
Total	**12875 (100%)**	**$ 8,595,332,400**



Spatial Analysis at County Level – Hudson County Example

Combined Scenario – 5ft inundation above MHHW + FEMA 500 year flood

Municipalities	Flood Exposed Area (acres)	Percent Flood Exposed Area	Total Annual Property Tax ($)	Flood Exposed Annual Property Tax ($)	Percent Flood Exposed Property Tax
Bayonne City	3169.32	64%	$ 188,669,612	$ 27,716,086	15%
East Newark Boro	32.89	45%	$ 4,142,255	$ 180,008	4%
Guttenberg Town	8.02	6%	$ 31,879,564	$ 2,625,885	8%
Harrison Town	464.55	55%	$ 34,810,383	$ 5,583,635	16%
Hoboken City	649.98	81%	$ 211,731,287	$ 136,087,251	64%
Jersey City	5599.87	55%	$ 813,111,689	$ 301,271,589	37%
Kearny Town	4618.44	71%	$ 152,217,613	$ 49,222,233	32%
North Bergen Twp	1476.07	44%	$ 140,782,126	$ 17,740,436	13%
Secaucus Town	3190.04	76%	$ 92,394,615	$ 56,951,491	62%
Union City	4.83	1%	$ 121,309,170	$ 2,941	0%
Weehawken Twp	191.84	38%	$ 41,031,106	$ 10,754,171	26%
West New York Town	77.62	12%	$ 94,060,777	$ 1,514,349	2%
County Total	**19483.46**	**59%**	**$ 1,926,140,198**	**$ 609,650,076**	**32%**



- The combined flooding scenario exposes about 60% of Hudson county's acreage and about one-third of the county's total annual property tax revenue.
- Individual community flood exposures vary. Example – percent flood exposed property tax varies from 64% in Hoboken to 0% in Union City.

New Jersey County and Statewide Summary Analysis of Environmental Flooding Scenarios by 21 Counties

Flooding Spatial Exposure (Acreage & Number of Parcels) for combined flooding scenario)

Parcels Flooded more than 50% either by 5 ft + FEMA 500

- NJ Flooding Impacts:
  - 1,291,037 Acres (26% of the state total) are exposed to flooding
  - 355,100 Parcels (12% of the state total) are exposed to flooding.
- Most spatially impacted counties: Atlantic, Cape May, Hudson, Ocean, and Salem.

County	Acres			Parcels		
	Total Area (Acres)	Flood Exposed Area (acres)	Percent Flood Exposed Area	Total Parcels	Flood Exposed Parcels	Percent Flood Exposed Parcels
Atlantic	390,814	139,125	36%	127,271	37,286	29%
Bergen	153,490	48,666	32%	279,961	19,007	7%
Burlington	524,901	101,402	19%	173,037	13,184	8%
Camden	145,598	19,105	13%	190,048	11,828	6%
Cape May	183,126	103,923	57%	95,194	61,605	65%
Cumberland	321,149	122,345	38%	61,934	5,675	9%
Essex	83,035	32,404	39%	174,593	8,409	5%
Gloucester	215,072	69,107	32%	112,067	5,054	5%
Hudson	32,982	19,483	59%	115,188	22,417	19%
Hunterdon	279,878	24,292	9%	52,000	1,507	3%
Mercer	146,442	48,097	33%	125,725	4,378	3%
Middlesex	202,859	41,698	21%	240,314	7,897	3%
Monmouth	310,833	50,673	16%	229,718	26,416	11%
Morris	308,083	63,180	21%	176,124	7,868	4%
Ocean	485,077	187,550	39%	297,000	86,014	29%
Passaic	127,008	19,950	16%	125,854	9,225	7%
Salem	222,158	68,287	31%	31,276	10,359	33%
Somerset	195,127	27,101	14%	115,876	3,455	3%
Sussex	342,617	28,942	8%	74,238	930	1%
Union	67,438	19,713	29%	147,789	11,085	8%
Warren	232,042	55,992	24%	46,402	1,501	3%
State Total	**4,969,728**	**1,291,037**	**26%**	**2,991,609**	**355,100**	**12%**

Flooding *Financial Exposure ($ market value and annual property tax payments)* for the combined flooding scenario

Parcels Flooded more than 50% either by 5 ft + FEMA 500

- Flooding financial exposure:
  - $243 billion of Property value (16% of $1.5 trillion State Total).
  - $4 Billion (12% of State $35 billion total) in Annual Property Tax Payments.
- Most financially exposed counties – Annual Property Tax Revenue:
  Atlantic, Bergen, Cape May, Hudson, Ocean and Salem.

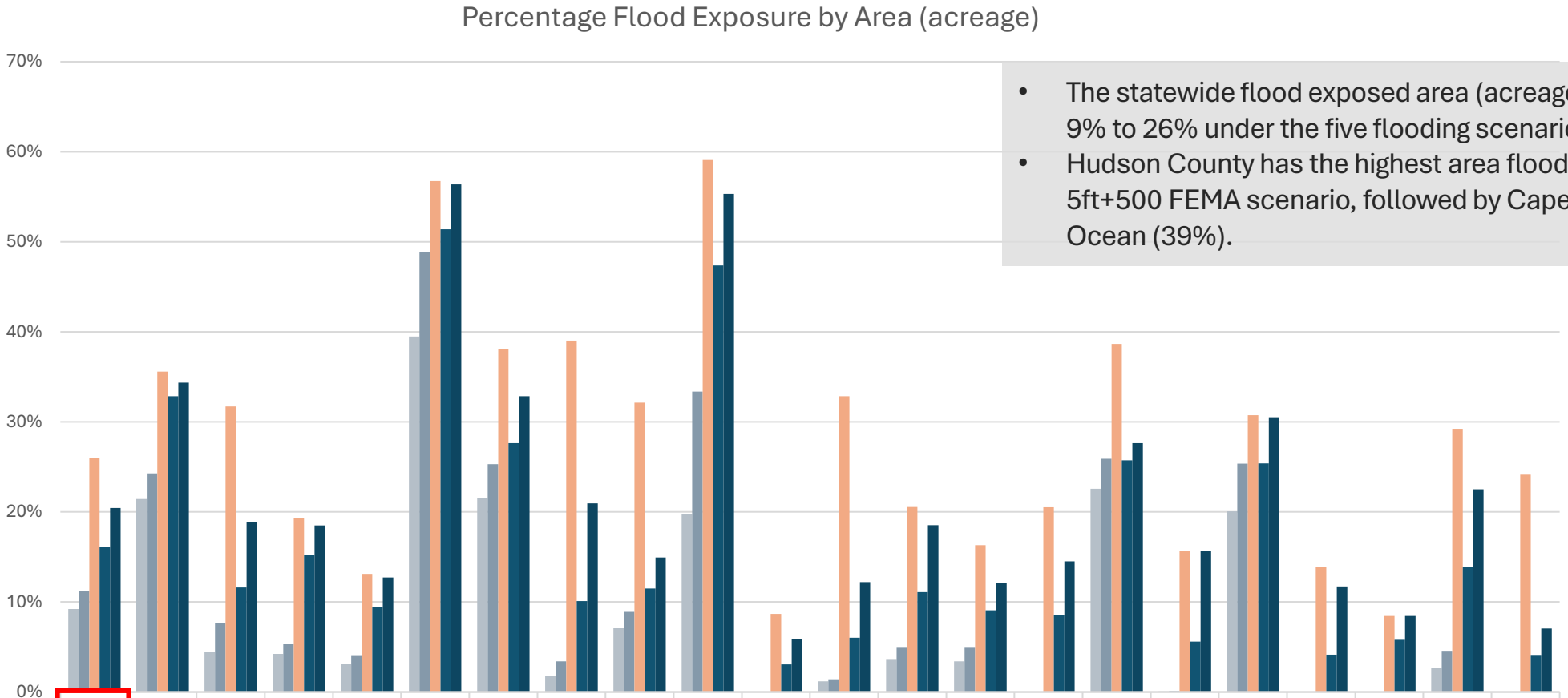
County	Property Value			Property Tax		
	Total Market Value	Flood Exposed Market Value	Percent Flood Exposed Market Value	Total Annual Property Tax	Flood Exposed Annual Property Tax	Percent Flood Exposed Property Tax
Atlantic	$ 37,860,090,100	$ 18,179,658,019	48%	$ 983,035,504	$ 394,670,402	40%
Bergen	$ 209,938,601,865	$ 20,733,699,507	10%	$ 4,610,141,009	$ 446,877,240	10%
Burlington	$ 58,226,922,542	$ 2,985,145,337	5%	$ 1,494,721,811	$ 87,527,989	6%
Camden	$ 50,028,619,068	$ 2,266,987,300	5%	$ 1,670,770,242	$ 72,073,093	4%
Cape May	$ 61,306,055,683	$ 50,476,711,049	82%	$ 563,249,017	$ 405,231,709	72%
Cumberland	$ 11,098,678,347	$ 589,972,883	5%	$ 333,151,561	$ 17,427,915	5%
Essex	$ 111,287,295,443	$ 8,063,554,910	7%	$ 3,143,816,941	$ 209,804,008	7%
Gloucester	$ 32,205,117,039	$ 1,454,134,950	5%	$ 979,023,383	$ 41,852,973	4%
Hudson	$ 105,427,564,430	$ 36,792,011,832	35%	$ 1,926,140,198	$ 609,650,076	32%
Hunterdon	$ 21,511,483,225	$ 625,382,777	3%	$ 536,679,702	$ 15,465,691	3%
Mercer	$ 57,753,907,688	$ 2,010,452,832	3%	$ 1,635,640,257	$ 68,991,181	4%
Middlesex	$ 137,626,786,512	$ 4,511,737,414	3%	$ 3,185,215,812	$ 107,603,906	3%
Monmouth	$ 144,078,164,939	$ 16,718,652,321	12%	$ 2,509,210,660	$ 261,855,325	10%
Morris	$ 107,864,711,543	$ 4,882,205,519	5%	$ 2,391,775,808	$ 103,069,287	4%
Ocean	$ 131,616,644,772	$ 55,606,846,870	42%	$ 2,182,438,394	$ 745,000,528	34%
Passaic	$ 64,158,106,264	$ 5,459,523,991	9%	$ 1,803,338,452	$ 152,801,723	8%
Salem	$ 5,676,240,984	$ 1,660,454,969	29%	$ 198,810,341	$ 66,735,039	34%
Somerset	$ 69,004,835,805	$ 1,558,475,777	2%	$ 1,454,263,674	$ 37,582,887	3%
Sussex	$ 20,180,341,377	$ 182,513,645	1%	$ 585,212,890	$ 5,158,688	1%
Union	$ 100,692,975,464	$ 8,371,883,489	8%	$ 2,516,969,232	$ 220,169,383	9%
Warren	$ 12,270,071,346	$ 282,746,603	2%	$ 358,870,547	$ 8,460,751	2%
State Total	**$ 1,549,813,214,436**	**$ 243,412,751,995**	**16%**	**$ 35,062,475,436**	**$ 4,078,009,792**	**12%**

Detailed Analysis

The following slides include the detailed state and county analyses by individual flooding exposure parameters

- Percentage flood exposure by acreage
- Percentage flood exposure by number of parcels
- Percentage of flood exposed market value
- Flood exposed annual property tax revenue

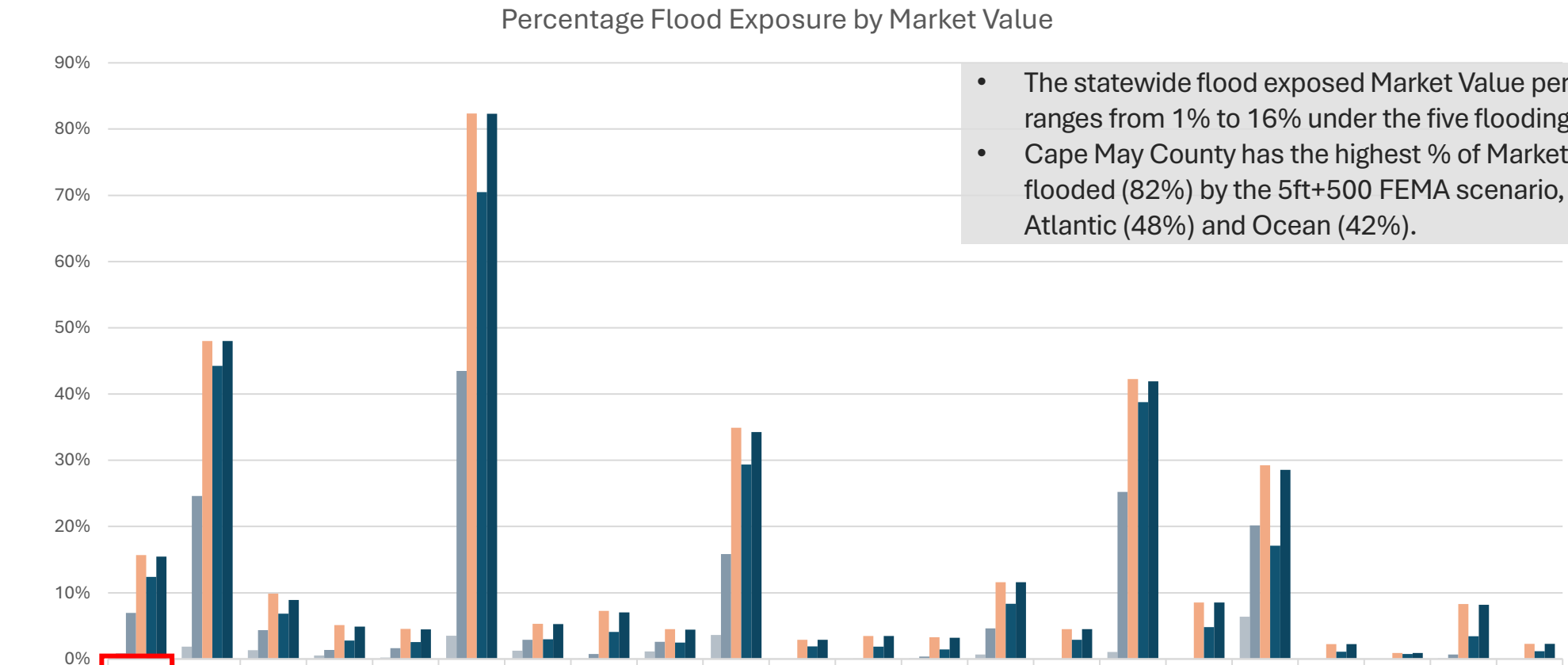
NJ flooding *Spatial exposure– Flooded percentage of Acreage* by state/county and five flooding scenarios



- The statewide flood exposed area (acreage) ranges from 9% to 26% under the five flooding scenarios.
- Hudson County has the highest area flooded (59%) by the 5ft+500 FEMA scenario, followed by Cape May (57%) and Ocean (39%).

	State Total	Atlantic	Bergen	Burlington	Camden	Cape May	Cumberland	Essex	Gloucester	Hudson	Hunterdon	Mercer	Middlesex	Monmouth	Morris	Ocean	Passaic	Salem	Somerset	Sussex	Union	Warren
2ft	9%	21%	4%	4%	3%	39%	22%	2%	7%	20%	0%	1%	4%	3%	0%	23%	0%	20%	0%	0%	3%	0%
5ft	11%	24%	8%	5%	4%	49%	25%	3%	9%	33%	0%	1%	5%	5%	0%	26%	0%	25%	0%	0%	5%	0%
5ft+500yr	26%	36%	32%	19%	13%	57%	38%	39%	32%	59%	9%	33%	21%	16%	21%	39%	16%	31%	14%	8%	29%	24%
100yr	16%	33%	12%	15%	9%	51%	28%	10%	12%	47%	3%	6%	11%	9%	9%	26%	6%	25%	4%	6%	14%	4%
500yr	20%	34%	19%	19%	13%	56%	33%	21%	15%	55%	6%	12%	19%	12%	15%	28%	16%	31%	12%	8%	23%	7%

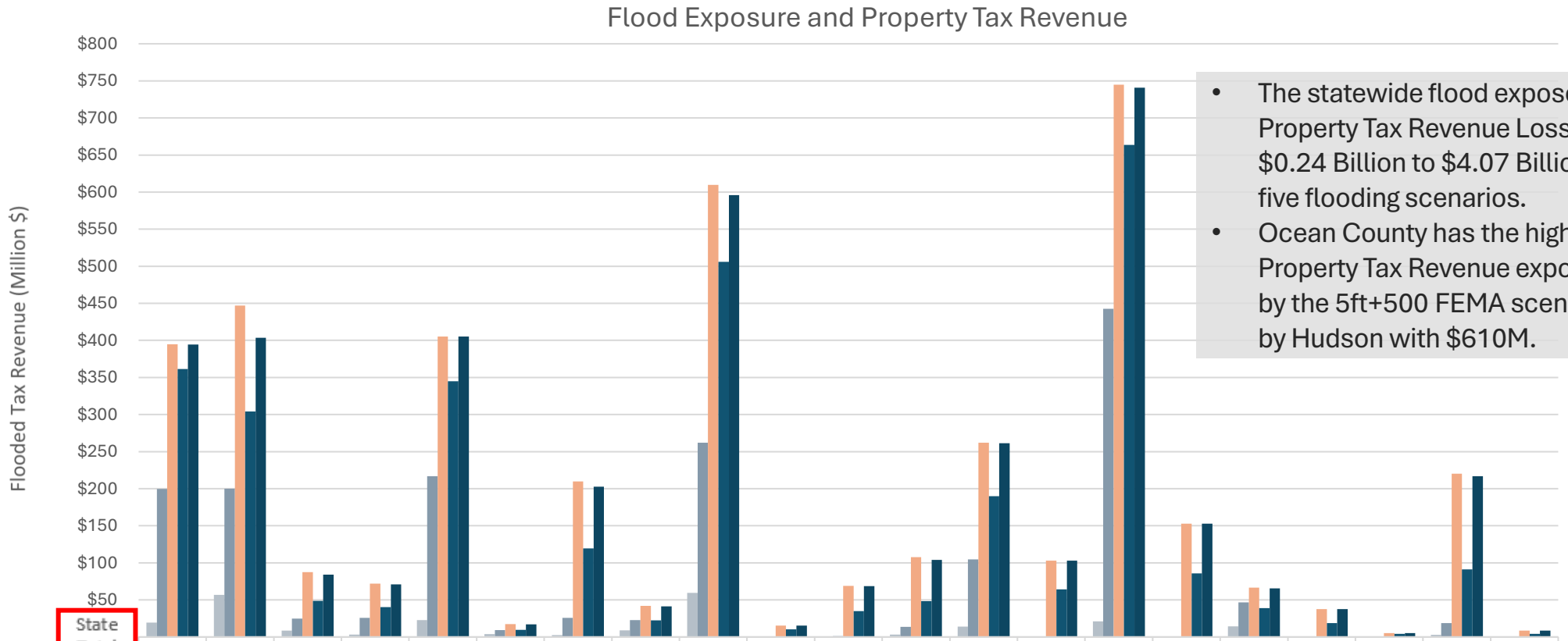
NJ flooding *financial exposure – Percentage flooded Market value by* state/county and five flooding scenarios



	State Total	Atlantic	Bergen	Burlington	Camden	Cape May	Cumberland	Essex	Gloucester	Hudson	Hunterdon	Mercer	Middlesex	Monmouth	Morris	Ocean	Passaic	Salem	Somerset	Sussex	Union	Warren
2ft	1%	2%	1%	1%	0%	3%	1%	0%	1%	4%	0%	0%	0%	1%	0%	1%	0%	6%	0%	0%	0%	0%
5ft	7%	25%	4%	1%	2%	43%	3%	1%	3%	16%	0%	0%	0%	5%	0%	25%	0%	20%	0%	0%	1%	0%
5ft_500yr	16%	48%	10%	5%	5%	82%	5%	7%	5%	35%	3%	3%	3%	12%	5%	42%	9%	29%	2%	1%	8%	2%
100yr	12%	44%	7%	3%	3%	70%	3%	4%	2%	29%	2%	2%	1%	8%	3%	39%	5%	17%	1%	1%	3%	1%
500yr	15%	48%	9%	5%	4%	82%	5%	7%	4%	34%	3%	3%	3%	12%	5%	42%	9%	29%	2%	1%	8%	2%

- The statewide flood exposed Market Value percentage ranges from 1% to 16% under the five flooding scenarios.
- Cape May County has the highest % of Market Value flooded (82%) by the 5ft+500 FEMA scenario, followed by Atlantic (48%) and Ocean (42%).

NJ flooding *financial exposure – Flooded annual property tax revenue impact (M$)* by state/county and five flooding scenarios



- The statewide flood exposed Annual Property Tax Revenue Loss ranges from $0.24 Billion to $4.07 Billion under the five flooding scenarios.
- Ocean County has the highest Annual Property Tax Revenue exposure $745M by the 5ft+500 FEMA scenario followed by Hudson with $610M.

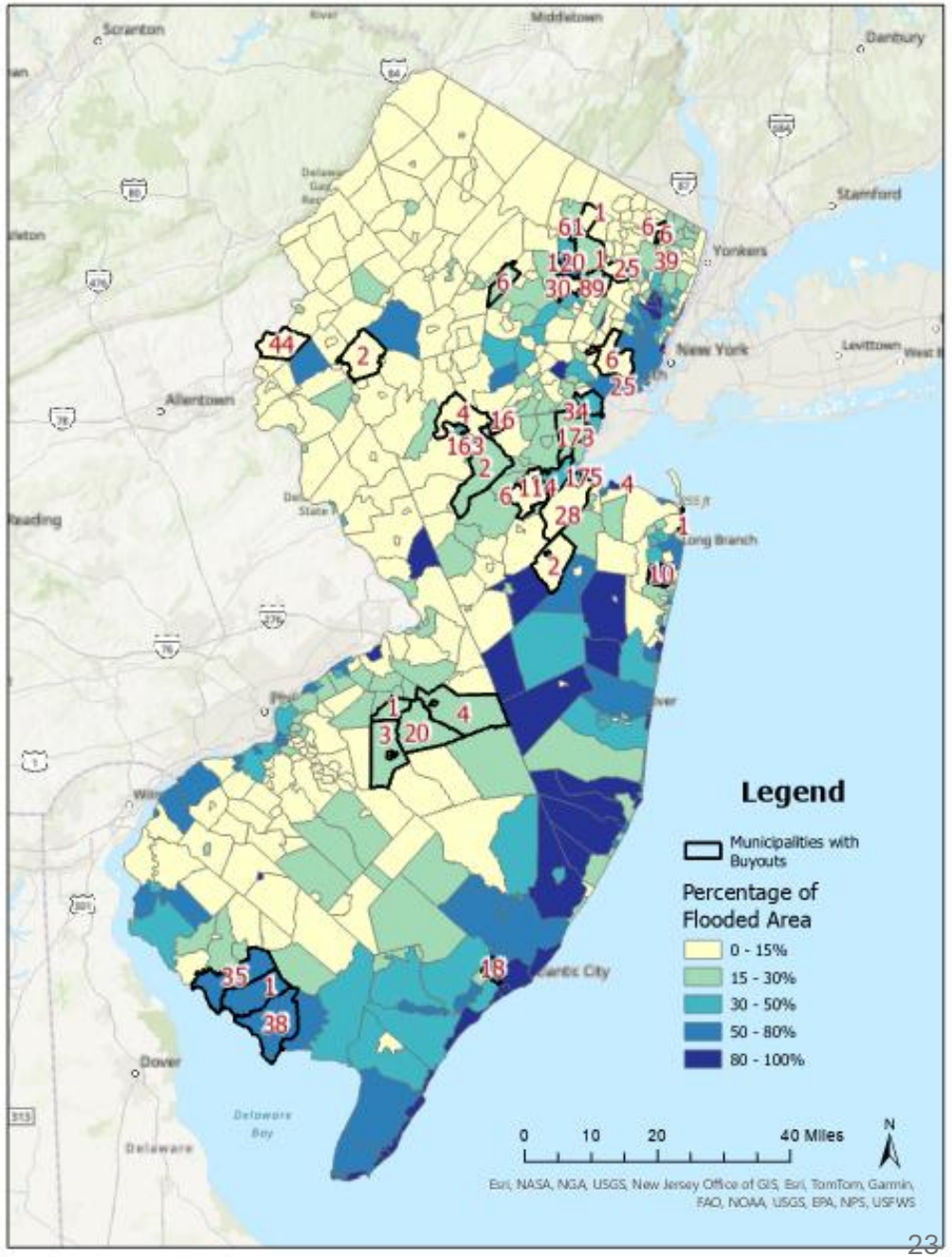
	State Total (Billion)	Atlantic	Bergen	Burlington	Camden	Cape May	Cumberland	Essex	Gloucester	Hudson	Hunterdon	Mercer	Middlesex	Monmouth	Morris	Ocean	Passaic	Salem	Somerset	Sussex	Union	Warren
2ft	$0.24	$19	$57	$9	$3	$23	$4	$3	$9	$60	$-	$1	$3	$14	$-	$21	$-	$14	$-	$-	$3	$-
5ft	$1.61	$200	$200	$25	$26	$217	$9	$26	$23	$262	$-	$1	$14	$105	$-	$443	$0	$47	$-	$-	$19	$-
5ft+500yr	$4.07	$395	$447	$88	$72	$405	$17	$210	$42	$610	$15	$69	$108	$262	$103	$745	$153	$67	$38	$5	$220	$8
100yr	$3.01	$361	$304	$49	$40	$345	$10	$119	$23	$506	$10	$35	$48	$190	$64	$664	$86	$39	$19	$4	$91	$4
500yr	$3.99	$395	$403	$84	$71	$405	$17	$203	$41	$596	$15	$69	$104	$261	$103	$741	$153	$65	$38	$5	$217	$8

Note - State impact in *$ billions* and *County* impact in *$ millions*

Intersection of NJ Flood Exposure to Statewide Property Buyouts, Construction Activity and Municipal Bond Ratings

Flood Exposure and Number of Buyouts

- The map depicts municipalities with more than 50% area exposed to flooding in **5ft + 500yr** scenario in darker blue colors.
- As the flood scenario takes into account both Coastal and Inland flooding, the flood exposure is spread throughout the State
- The municipalities with dark outline are where bouyots have taken place and the number depicts the number of buyout parcels.
- While NJ buyouts have been important in reducing flooding property hazards, this strategy mostly has not been implemented in the highest flood-exposed NJ municipalities (e.g., large swaths of the NJ shore).



Accomplishment, Limitations and Future Analytic Enhancements

Accomplishments

- First large-scale ***macro*** (statewide) quantification of the intersection of flooding challenge and public finance (property sales/tax).

- In tandem, the analysis provides ***micro*** (individual parcel) level quantification of the flooding exposure and public finance intersection.

- Analysis begins to explore the intersection of *flood exposure/public finance* to a policy response (buyouts) and other critical associations (e.g., construction activity and municipal bond rating).

- As the analysis is embedded in the larger suite of the NJ Adapt tools (e.g. public health) the *scope of the analysis can be extended*.

Limitations and future analytic enhancements

- Limitations
  - As all data systems/tools NJ Adapt has inherent quantification limitations, especially at individual parcel level.
  - Impact effects may be *temporary* and can be *mitigated by various policy responses* (e.g., property elevation)
- Future Enhancements
  - Refine impacts to differentiate temporary vs long term outcomes, and policy-mitigated impacts
  - Extend the impact metrics e.g., quantify flooding impacts to public/historic resources
  - Expand the mapping *of* the Impact metrics
  - Quantify Impacts at multiple geographic levels (e.g., individual community, all communities within a county or special planning area; as well as statewide)
  - Analyze greater/lesser flood challenge by *community socioeconomic* characteristics (e.g., race and poverty)
  - Analyze greater/lesser flood challenge by *community public finance characteristics* (e.g., property tax burden and municipal reliance on the property tax)